

Chapter 4-Part 1: Cell Structure

- Cells are the units of life
- What do all cells have in common?
- What are the different cell types & what makes each unique?
- How do cells stick together & communicate?

Corresponds with OpenStax Biology 2e Chapter 4

Cells are the units of life

- An animal, a plant, a mushroom, & bacteria seem very different, but all are made up of cells
 - **Cell** = the smallest unit of life
 - events that occur inside a cell make life possible

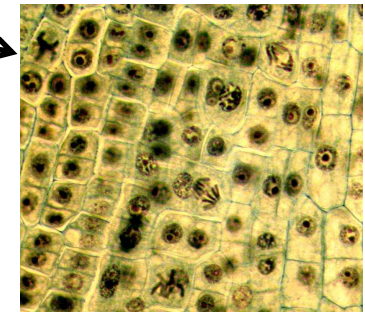
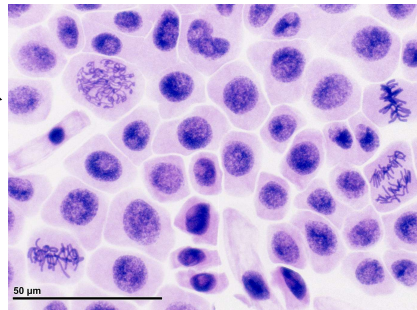


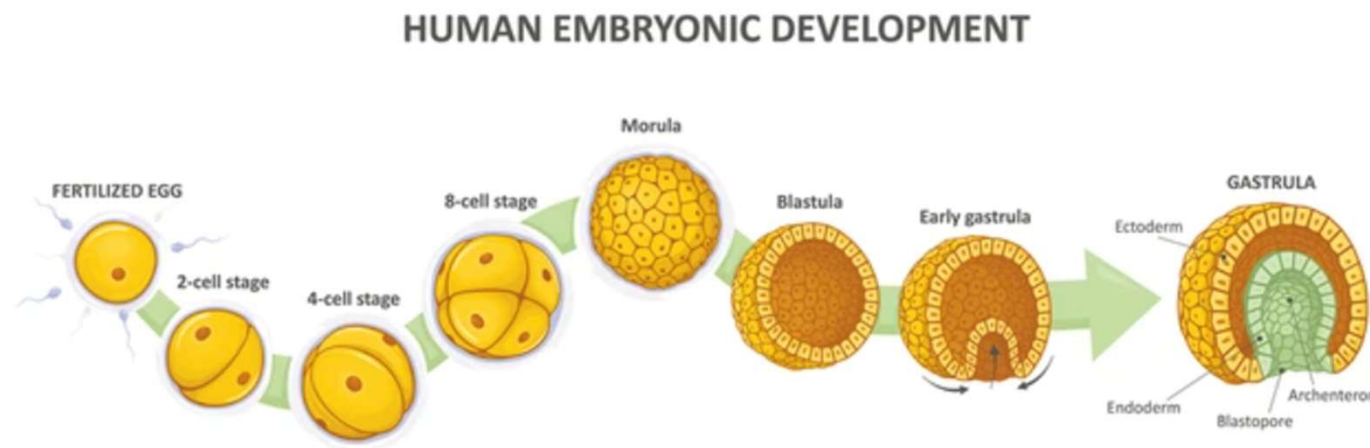
Figure: [https://commons.wikimedia.org/wiki/File:Mitosis_\(261_13\)_Pressed_root_meristem_of_onion_\(cells_in_prophase,_metaphase,_anaphase,_telophase\).jpg](https://commons.wikimedia.org/wiki/File:Mitosis_(261_13)_Pressed_root_meristem_of_onion_(cells_in_prophase,_metaphase,_anaphase,_telophase).jpg)

Figure: <https://www.flickr.com/photos/foundanimalsfoundation/8055195809>

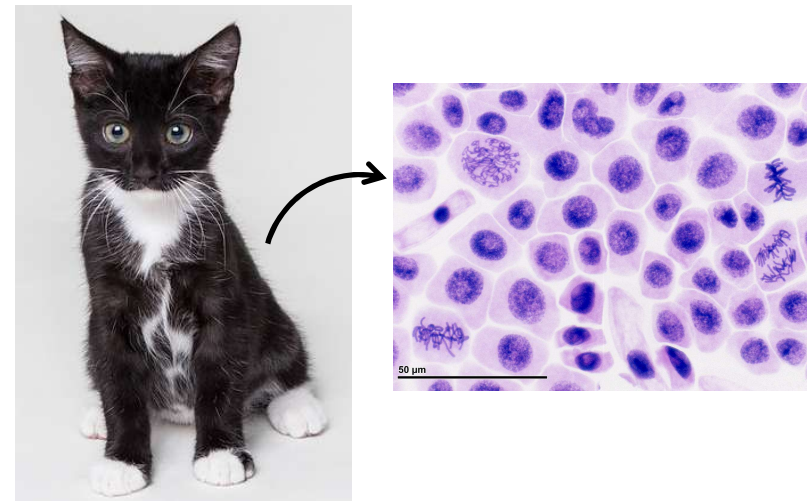
Figure: <https://www.flickr.com/photos/71183136@N08/6947182364>

Figure: https://commons.wikimedia.org/wiki/File:Uraro_plant.jpg

- The invention of microscopes allowed us to see that all living things were made entirely of cells
- Further studies showed that every cell came from another, pre-existing cell
 - e.g. an adult human is made up of trillions of cells, all of which came from just one cell: a single fertilized egg



- The facts that all living organisms are made up of one or more cells & that all cells arise from preexisting cells are the foundations of **cell theory**
 - 3 main principles of cell theory:
 - 1. all living things are made up of one or more cells
 - 2. the cell is the basic unit of life
 - 3. all cell come from preexisting cells



- The cell theory is still evolving to become more powerful, with additional modern ideas



Early cell theory

All organisms are made of one or more cells.

The cell is the fundamental unit of life.

All cells come from preexisting cells.

Remember
from chs 2 & 3

Additional ideas in modern cell theory

All cells have the same basic chemical composition.

All cells use energy.

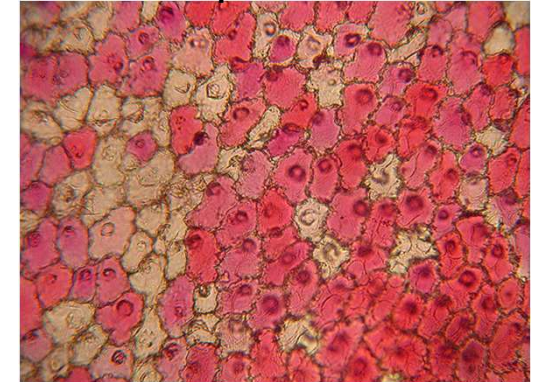
All cells contain DNA that is duplicated and passed on as each cell divides.

Coming
lectures

What do all cells have in common?

- Cells have a common characteristic: most are small
 - Bacteria, flea, whale, rose, oak...
all made of small cells
 - a “bigger” being has MORE cells, not bigger cells
 - Why?
 - Molecules must move in & out of a cell fast & efficiently: speed matters for life
 - not fast enough if cells are large

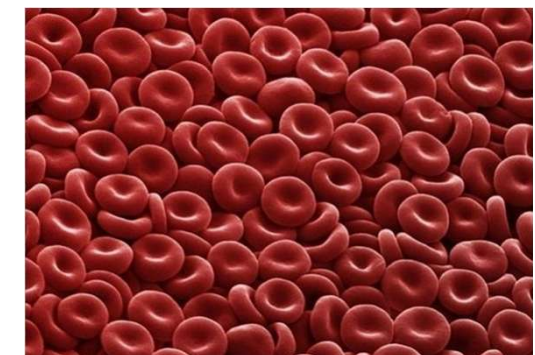
rose petal cells



bacterial cells



human blood cells



- Vast majority of cells are too small to see with the naked eye, but a few don't need to get things in & out: egg cells
 - egg cells can be large because they are filled with nutrients that nourish a growing embryo throughout development

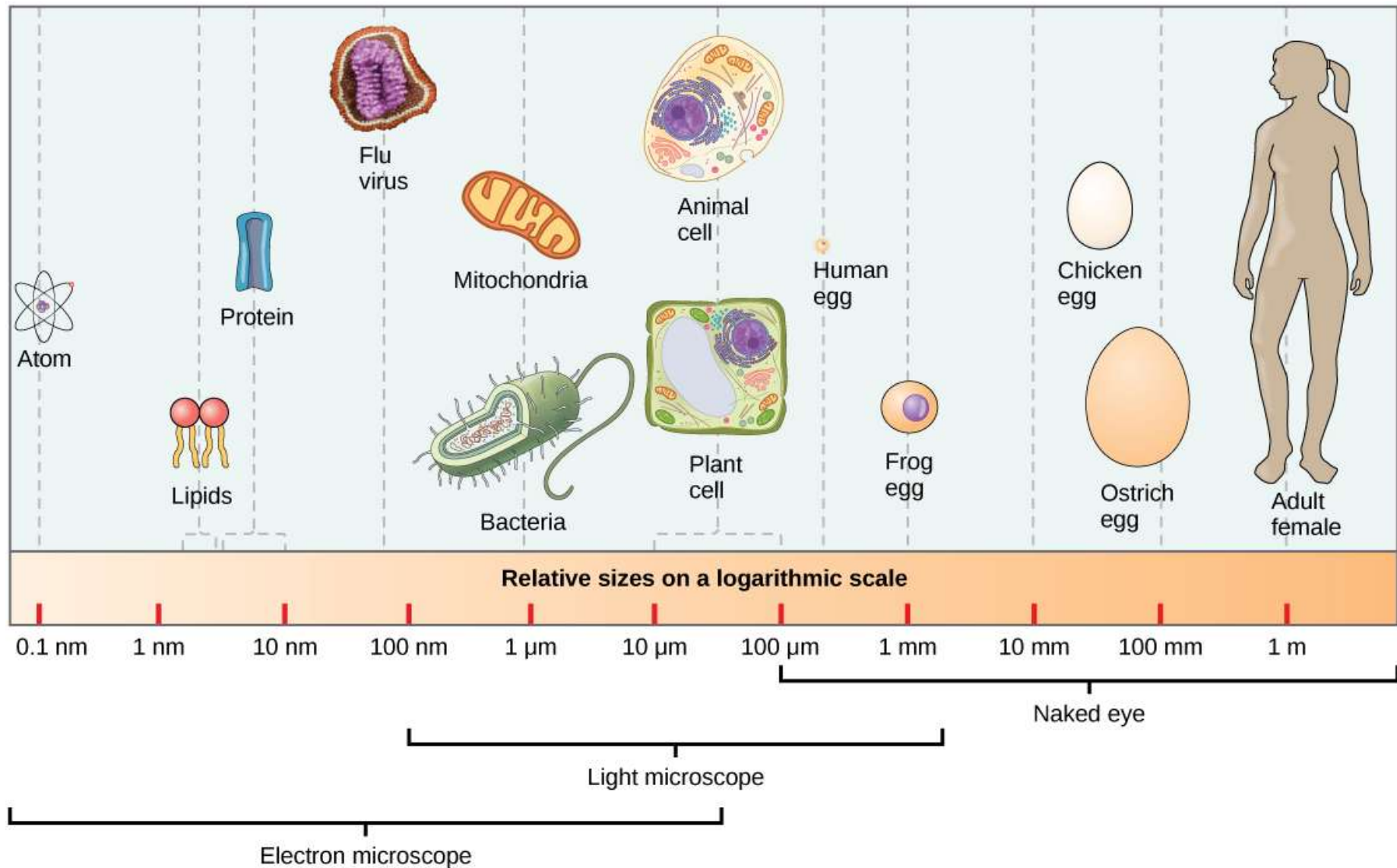
chicken
eggs



ostrich eggs can weigh 3lbs!



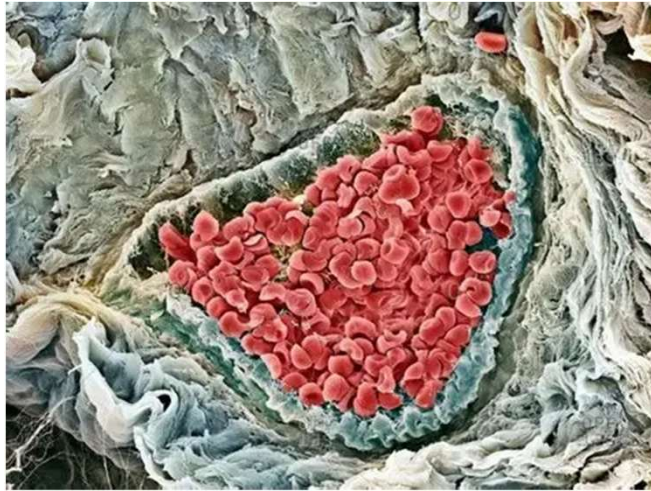
fish eggs (caviar)



This figure shows the relative sizes of microbes on a logarithmic scale (each unit of increase in a logarithmic scale represents a 10-fold increase in the quantity measured).

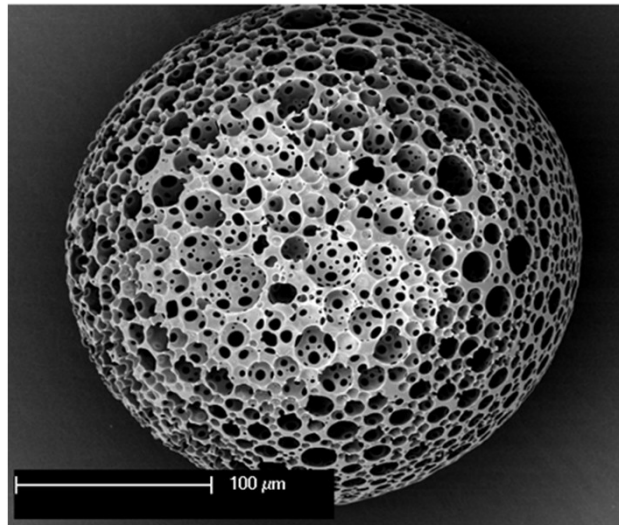
Electron Microscope

Magnify 1~50millions times!

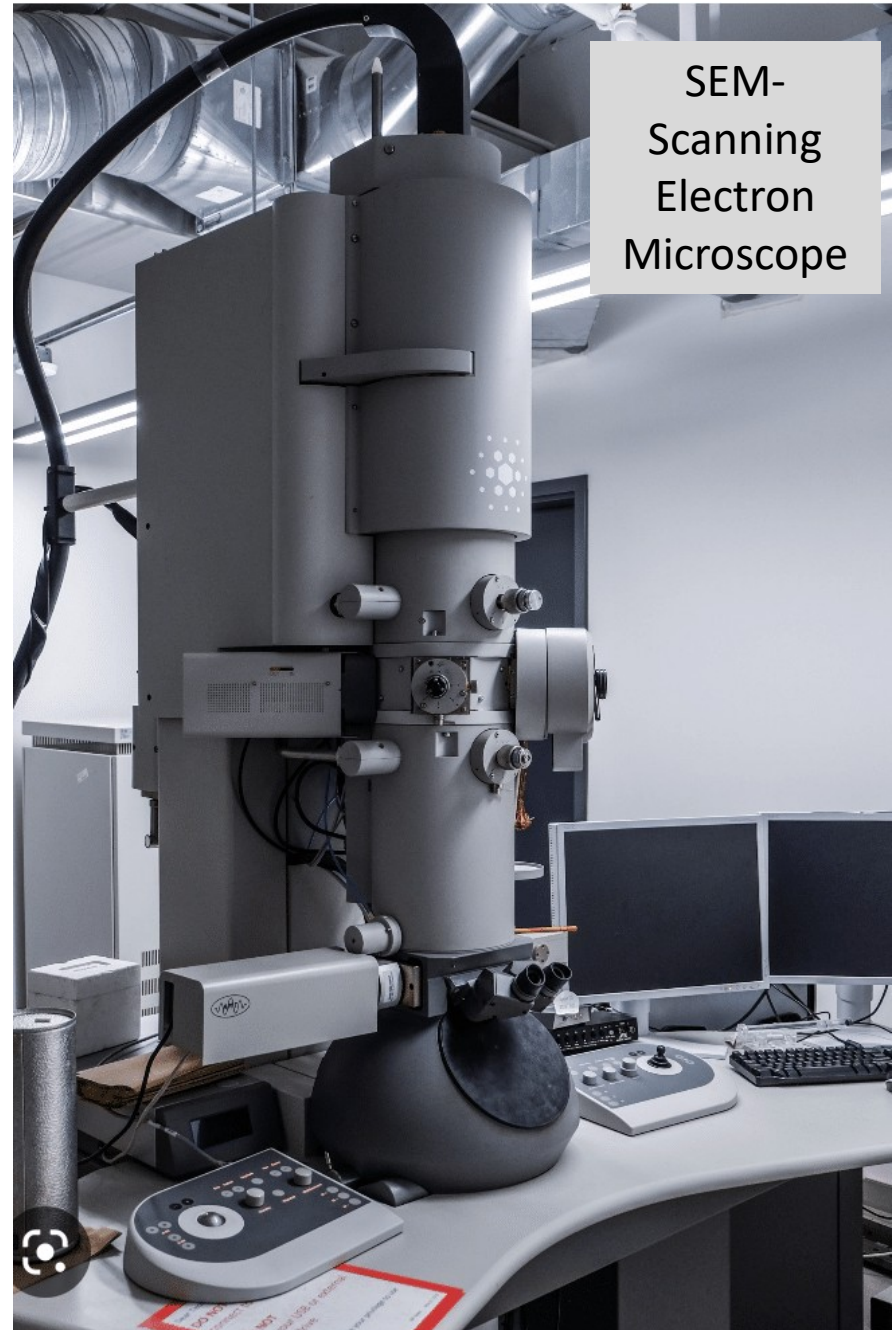


Red blood cell in a blood vessel, x650

<https://ar.inspiredpencil.com/pictures-2023/cell-wall-microscope>



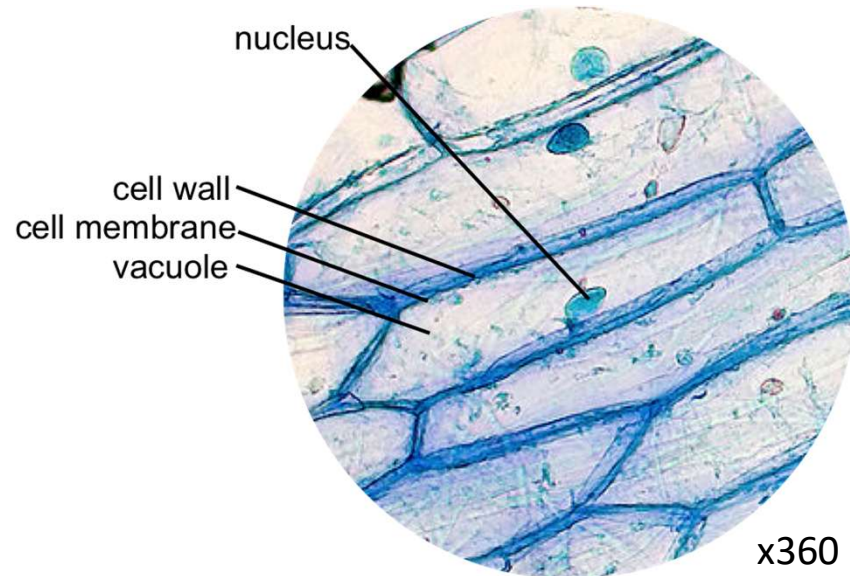
HIPE particles



SEM-
Scanning
Electron
Microscope

Light Microscope

Magnify 1~1000 times!



<https://ar.inspiredpencil.com/pictures-2023/cell-wall-microscope>



What do all cells have in common?

- Cells have many different parts (structures), each of which is responsible for a different function
- And different kinds of cells have different parts

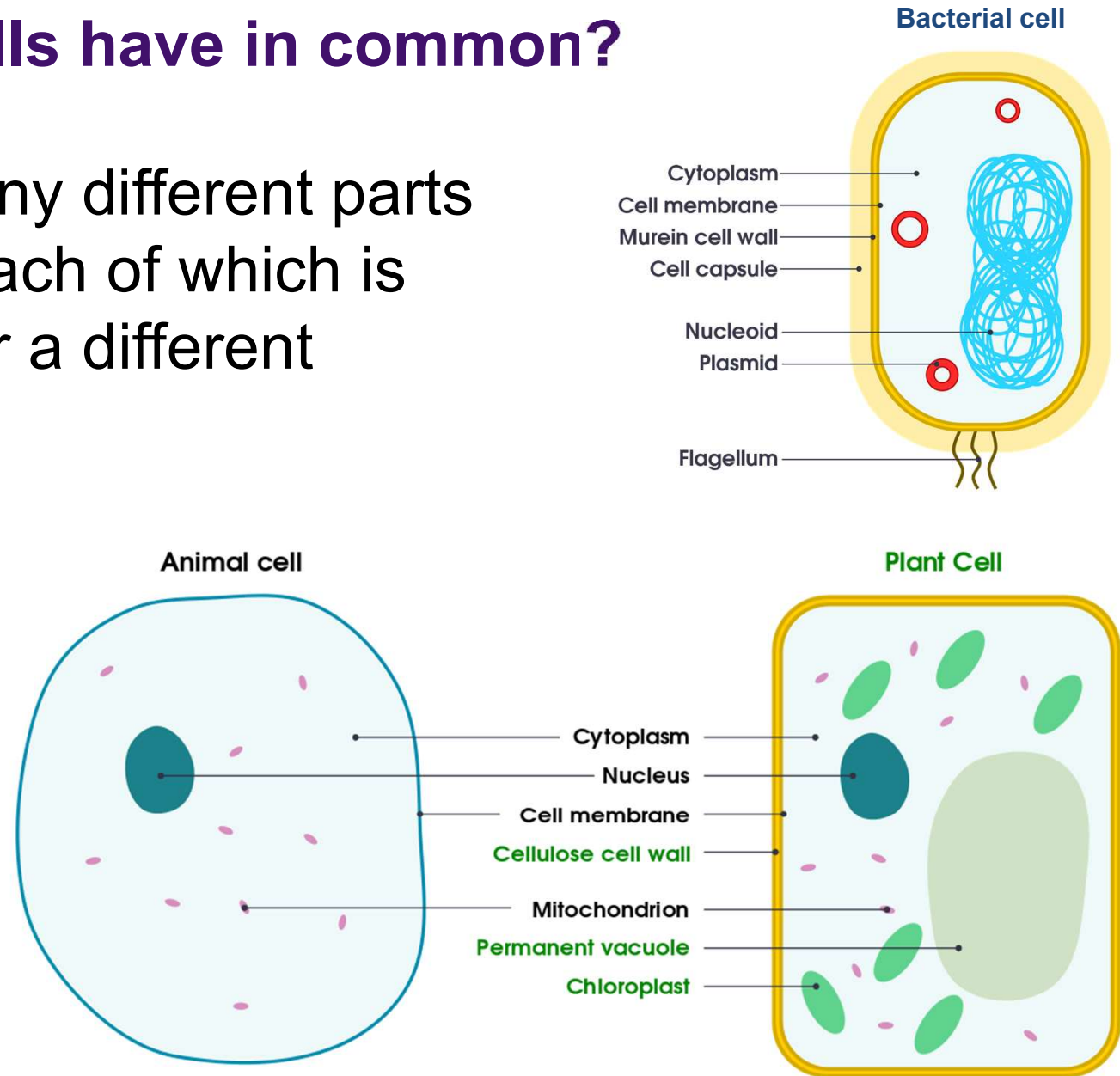
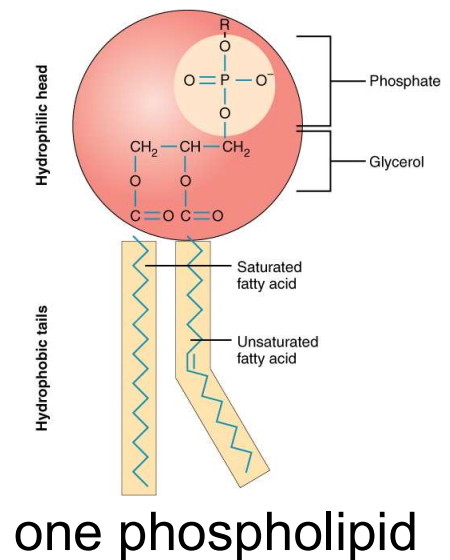
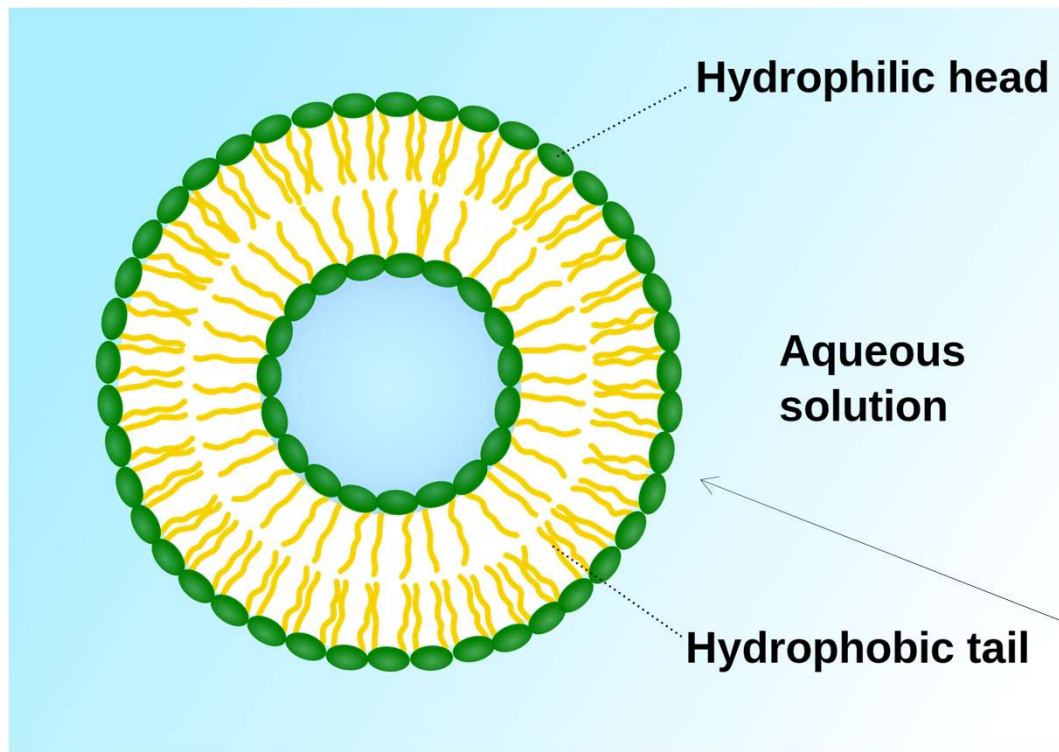


Figure: [https://commons.wikimedia.org/wiki/File:Differences_between_simple_animal_and_plant_cells_\(en\).svg](https://commons.wikimedia.org/wiki/File:Differences_between_simple_animal_and_plant_cells_(en).svg)

Figure: [https://commons.wikimedia.org/wiki/File:Simple_diagram_of_bacterium_\(en\).svg](https://commons.wikimedia.org/wiki/File:Simple_diagram_of_bacterium_(en).svg)

- However: ALL cells have the same 4 parts:
- 1. **Cell membrane**
 - encloses the cell & regulates what goes in & out
 - made of phospholipids



many phospholipids gather to create cell membranes

- **2. Cytoplasm** = all fluid & structures inside the cell membrane but outside of the nucleus
 - What about bacterial cells, which have no nucleus?
 - everything inside is cytoplasm!

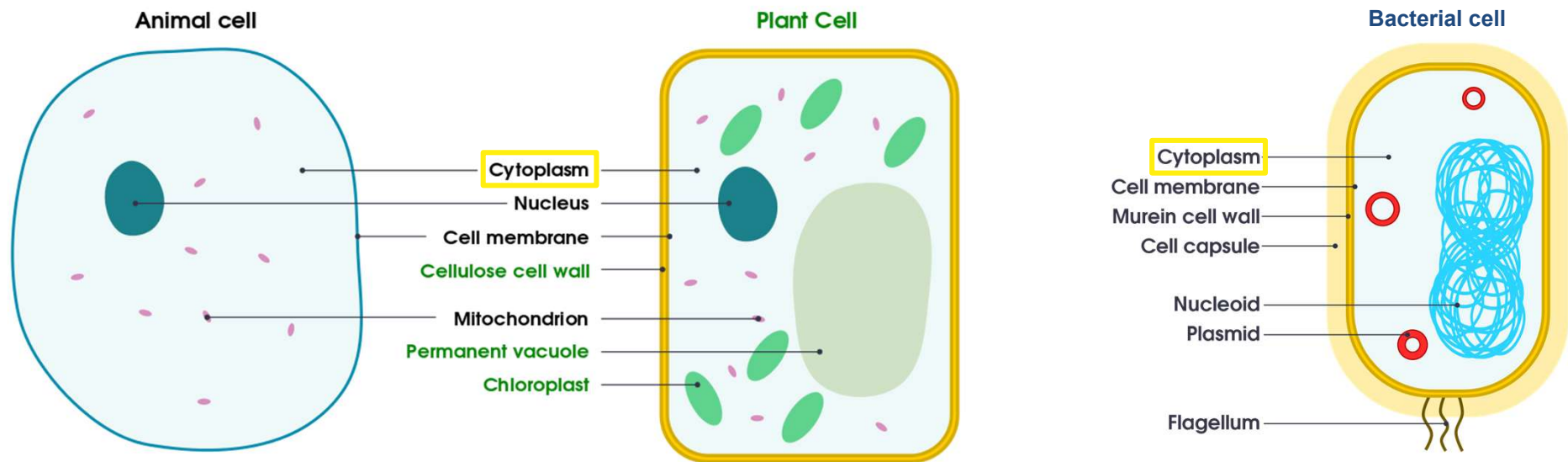
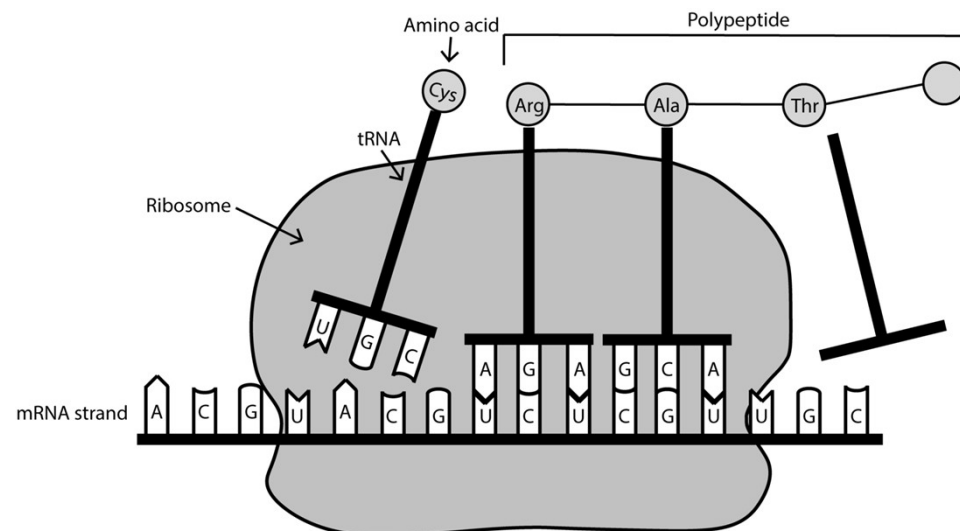


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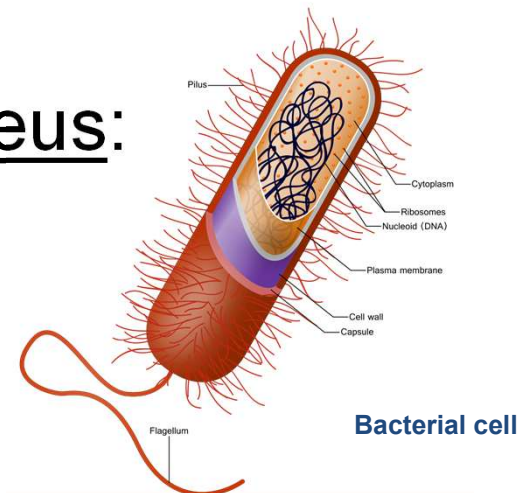
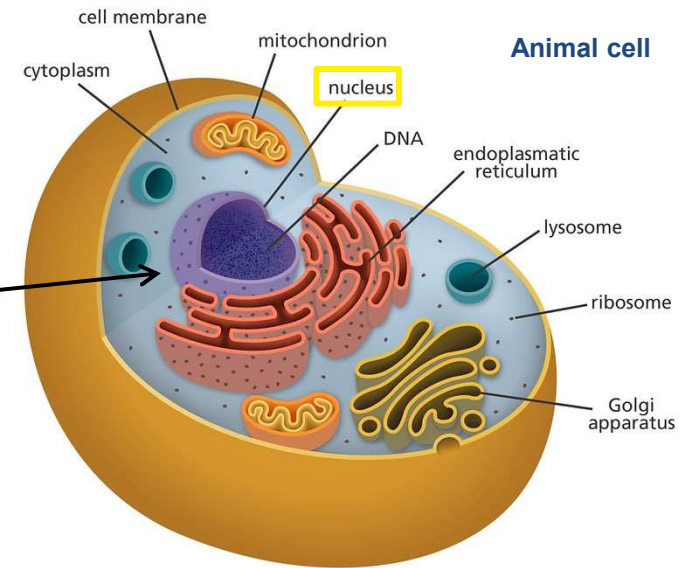
- **3. DNA, RNA, proteins (& other organic molecules)**
 - *Remember from ch 2: we need organic molecules to build & fuel our cells, which make up our bodies*
- **4. Ribosomes**
 - Make proteins (using directions from DNA & RNA)
 - since proteins do nearly everything, all living things (& their cells) must have ribosomes to build proteins

This is a diagram of a very important process for all cells that you'll learn about in chapter 7 – notice it involves proteins (polypeptides), RNA, and ribosomes, which are found in all cells.

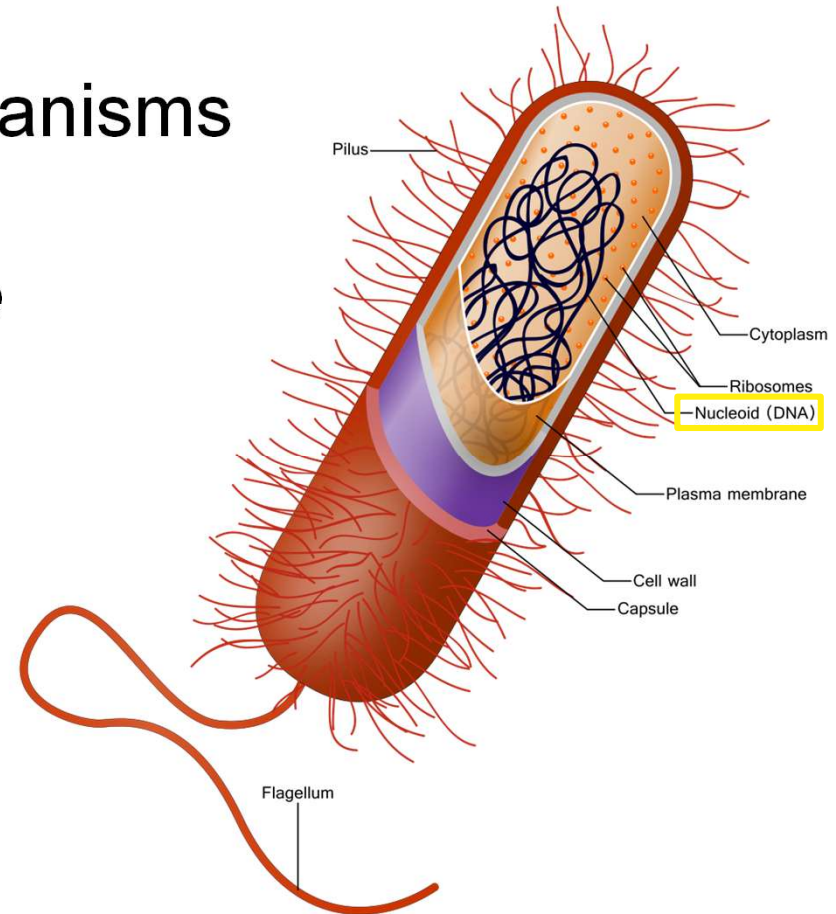


What are the different cell types & what makes each unique?

- There are 2 basic types of cells
 - Some have a nucleus:
eukaryotic cells
 - form bodies of animals, plants, fungi, & protists
 - Some do NOT have a nucleus:
prokaryotic cells
 - form bodies of bacteria & archaea

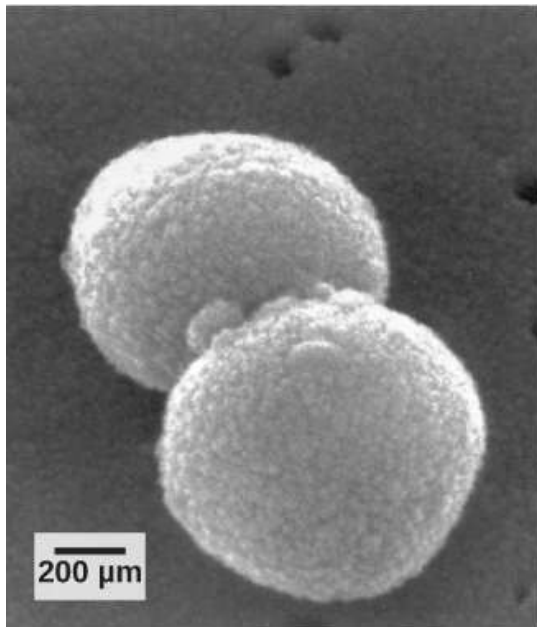


- **Prokaryotic cells** (prokaryotes)
 - smallest & most simple of cell types
 - have very few specialized structures inside
 - make up single-celled organisms
 - *can't see with the naked eye – need a microscope*
 - Instead of a nucleus, they have a **nucleoid region**: refers to where the DNA is located

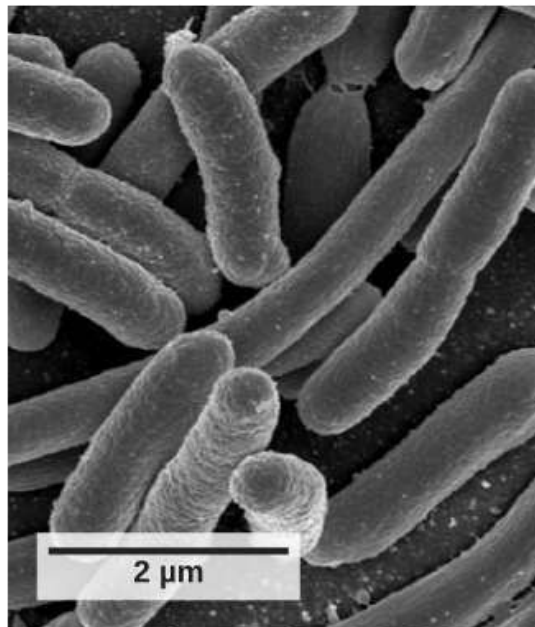


- Prokaryotic cells are so simple they usually take one of 3 common shapes
 - rod-shaped, spherical, or corkscrew-shaped

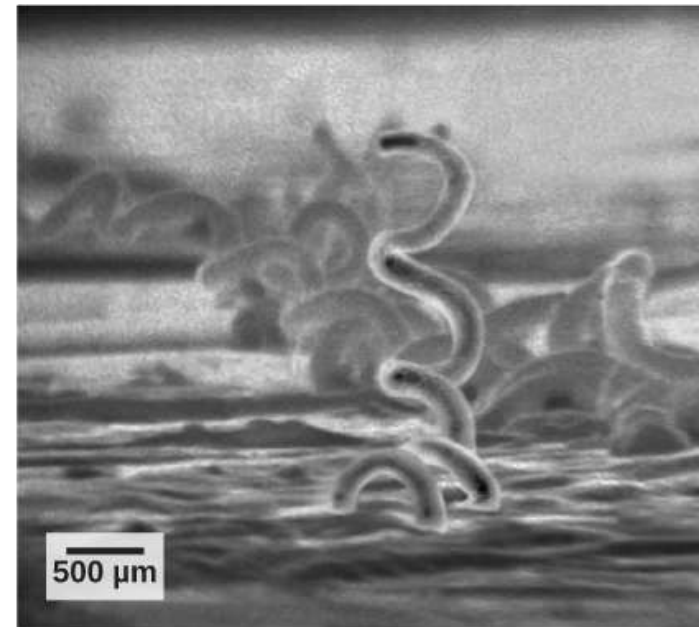
bacteria examples



spherical

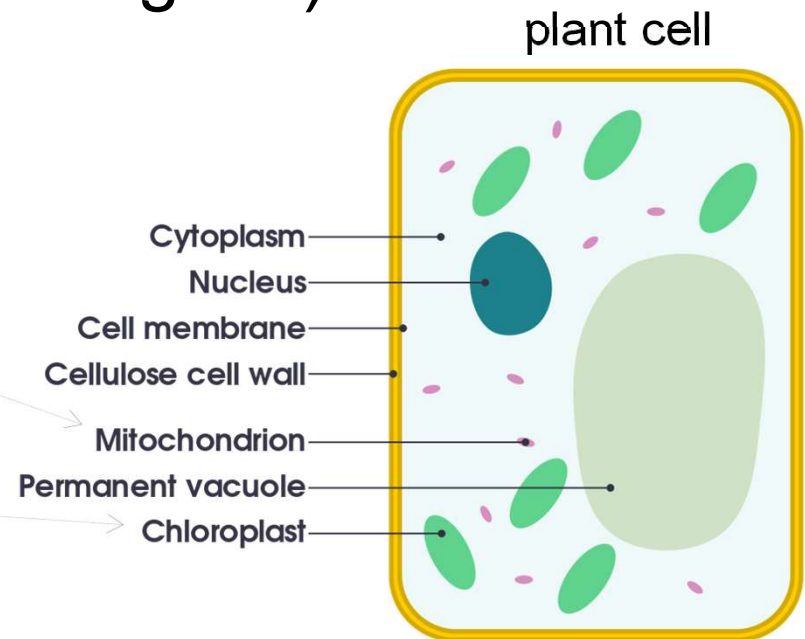


rod-shaped



corkscrew-shaped

- **Eukaryotic cells** (eukaryotes)
 - larger & more complex than prokaryotes
 - have a nucleus & many specialized structures called organelles
- **Organelles** = compartments that carry out specialized functions (little organs)
- Example organelles:
 - mitochondria generate energy
 - chloroplasts capture sunlight



- **Eukaryotic cells (eukaryotes)**

- make up all the multi-cellular organisms we can see (e.g. animals, plants, & fungi)
- make up some single-celled living things we can't see (e.g. protists & yeast – single-celled fungi)
 - our focus will be on plant & animal cells

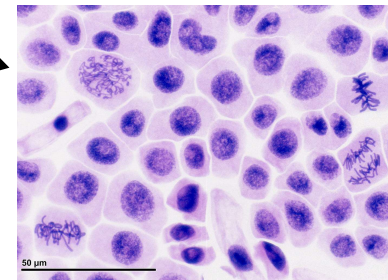
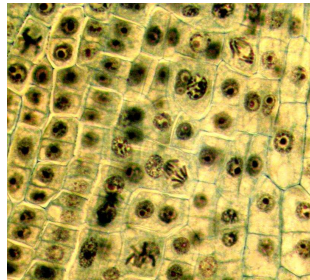


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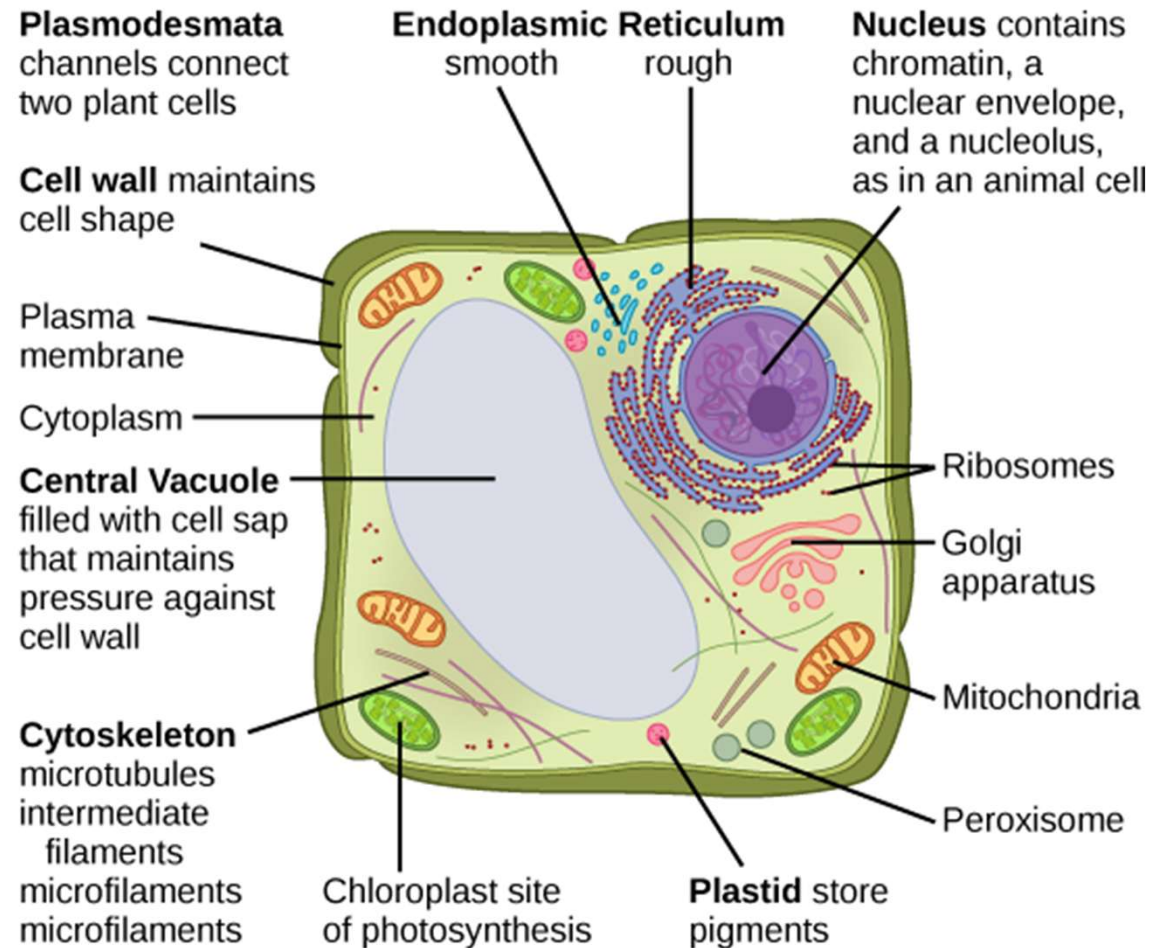
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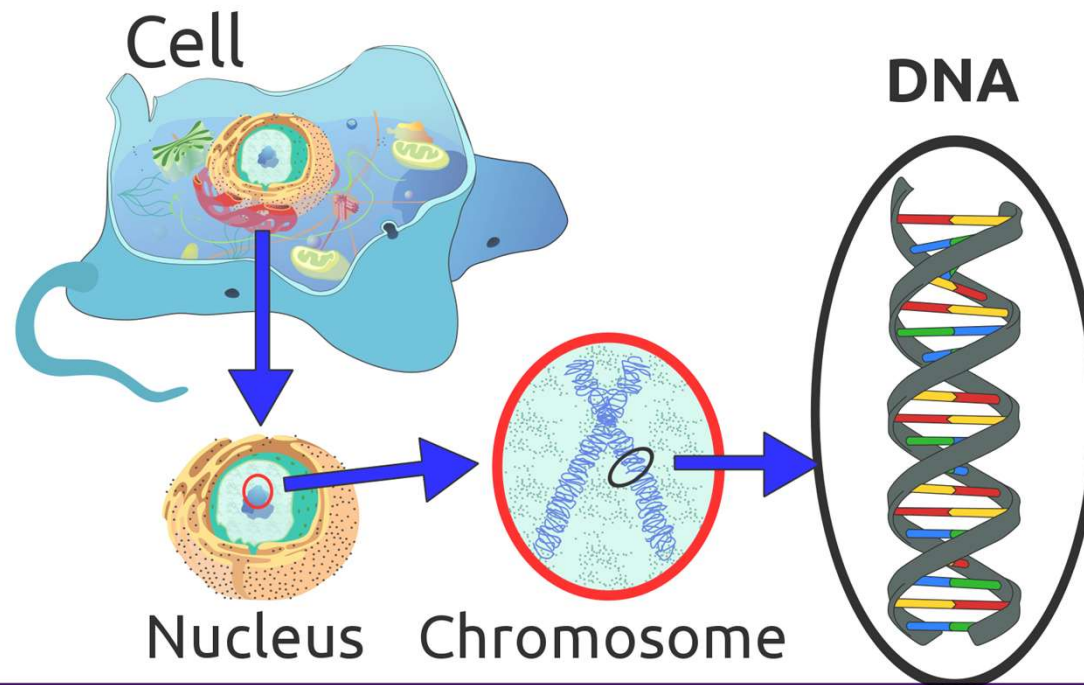
Figure: https://commons.wikimedia.org/wiki/File:Uraro_plant.jpg

- Although eukaryotic cells make up so many different living things, they all have similar structures

- So what structures do we find in eukaryotic cells & what do they do to keep us alive?



- **Nucleus** = control center of a eukaryotic cell
 - contains chromosomes, which are made of DNA
 - message in DNA leaves nucleus in the form of RNA & goes into *cytoplasm* to build proteins

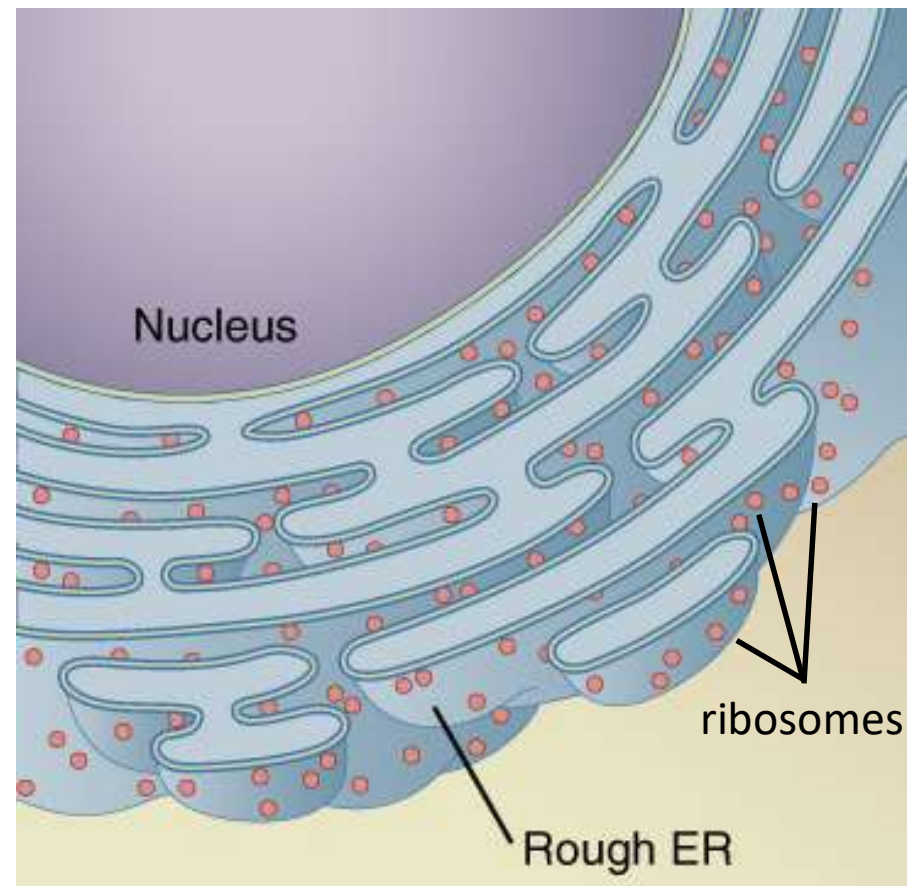


- **Ribosomes & rough endoplasmic reticulum (ER)**

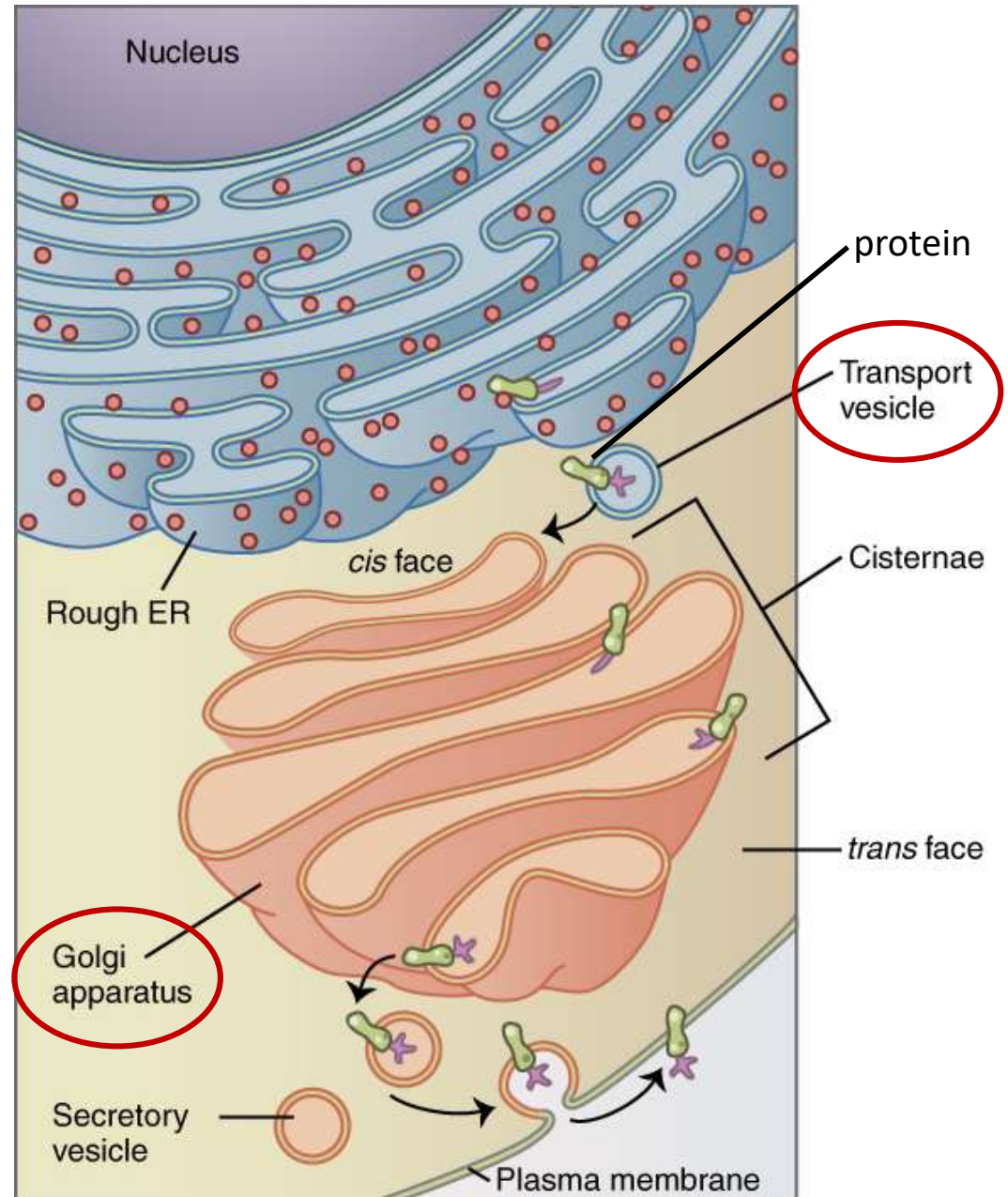
- Rough ER = “rough” because it is studded with ribosomes

- RNA leaves the nucleus & travels to ribosomes on the rough ER

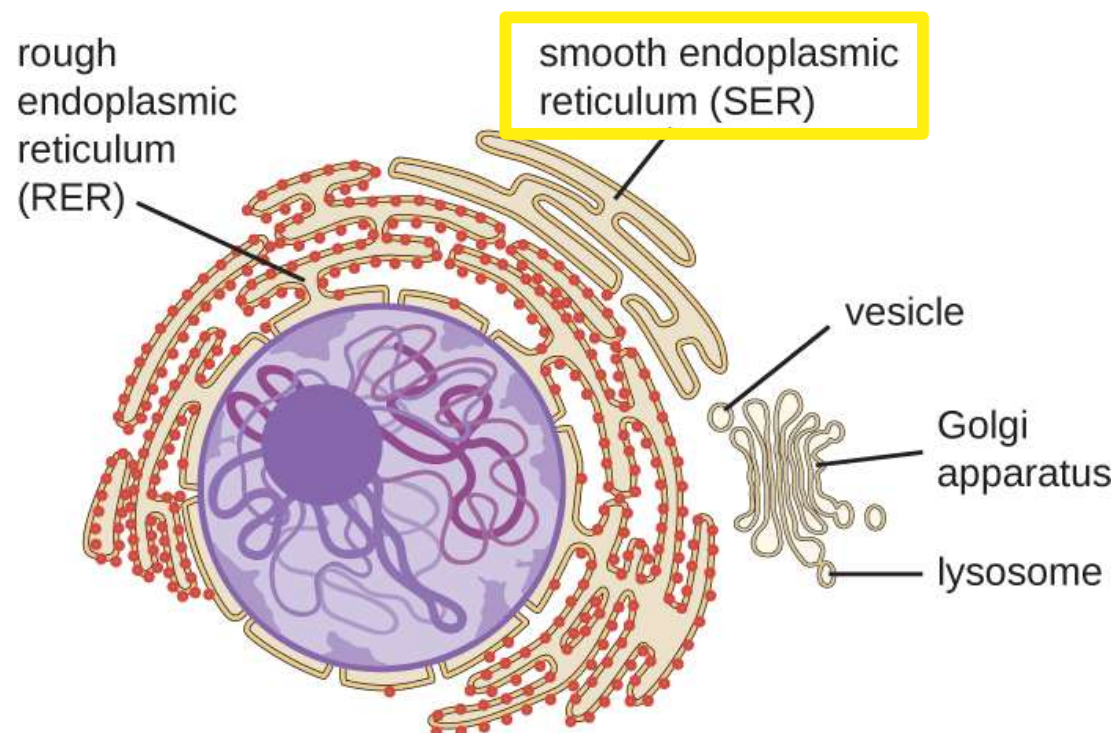
- makes proteins



- **Golgi apparatus:**
folds & packages
proteins, then sends
them to other parts
of the cell or out to
the blood stream
 - also packages
other molecules,
like lipids
 - packages are
called **vesicles**



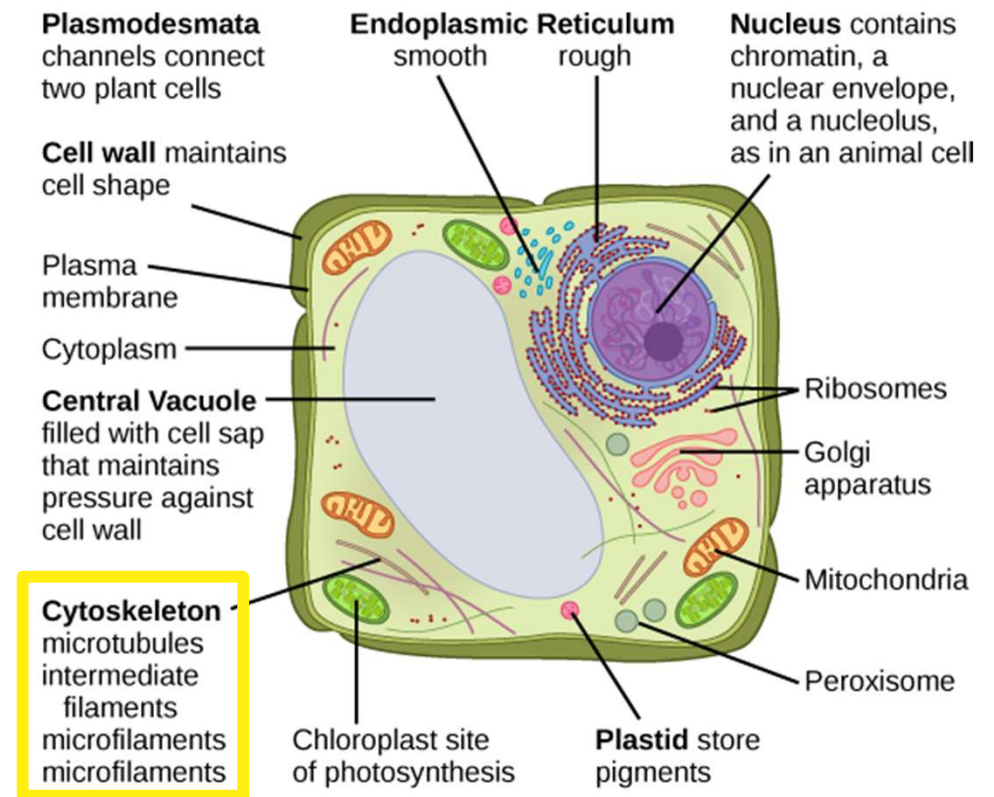
- **Smooth Endoplasmic Reticulum (ER)** makes lipids
 - sends the lipids through the Golgi apparatus to get sent out to other parts of the cell or to the blood stream
 - Smooth ER also detoxifies drugs & poisons



- How are packages sent around the cell without just floating around randomly?

– The cell uses a network similar to train tracks: the **cytoskeleton**

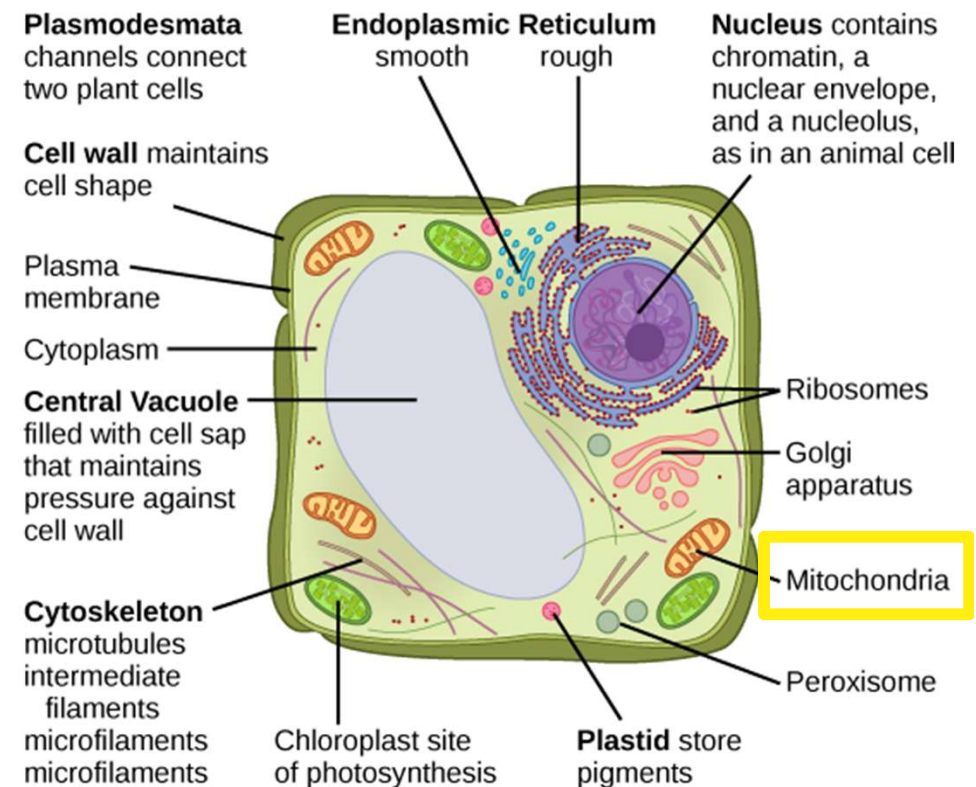
- gives a cell shape & support, but also allows for movement/ travel in the cell



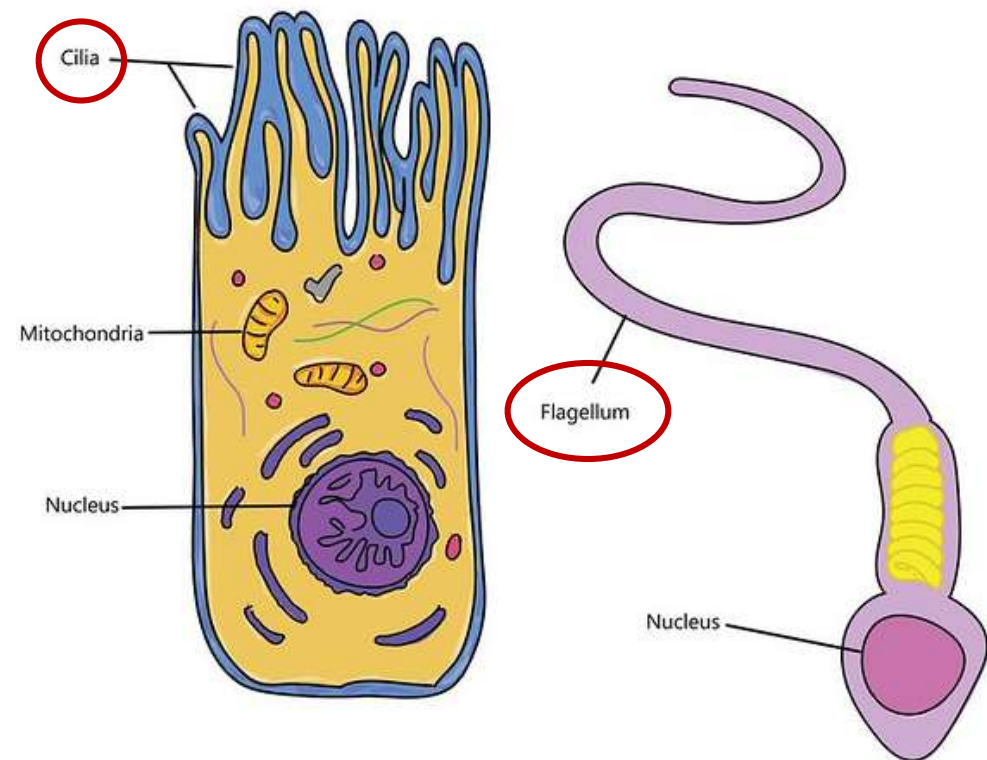
- Where do cells process energy? (to move things around on cytoskeleton & do all things to stay alive)

– **Mitochondria** = “powerhouse of the cell”

- breaks apart glucose (mono-saccharides) to provide energy to the cell
 - *this process is called “glucose breakdown” - more on this in a future chapter*

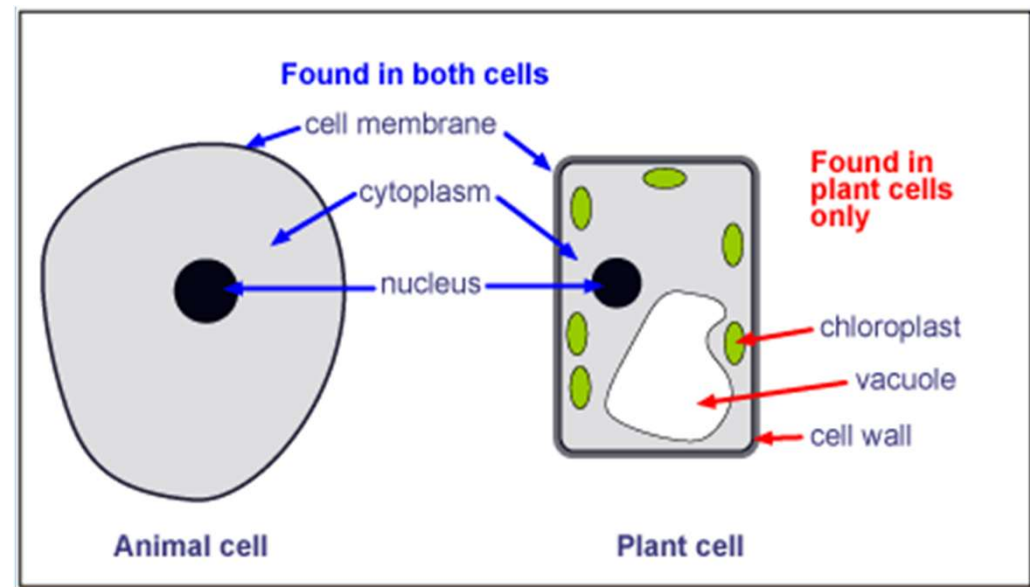


- Do animal cells have anything that plant cells don't?
 - Some animal cells have cilia or flagella
 - **Cilia** = short, numerous extensions of the membrane that move fluid past a cell
 - **Flagella** = long tails that propel cells, allowing for locomotion



- Do plant cells have anything that animal cells don't?
 - Plants have a cell wall, chloroplasts, & a central vacuole
 - **Cell wall** = coating over/outside the cell membrane that protects & supports cells

- *more on what helps support animal cells later*

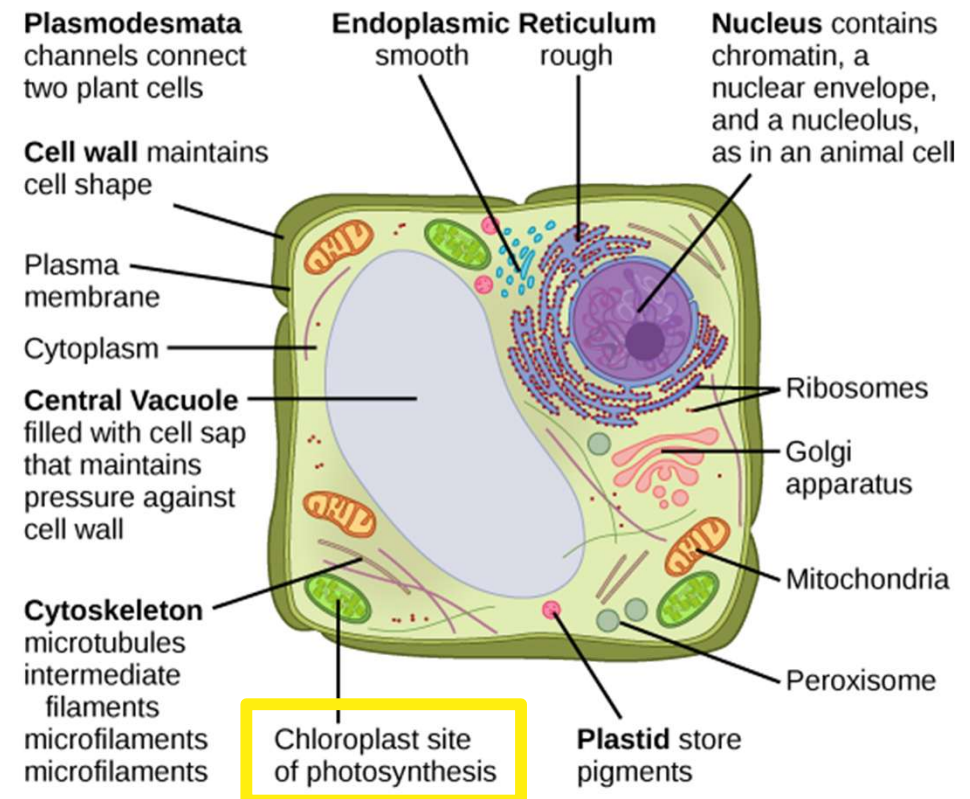


- Do plant cells have anything that animal cells don't?

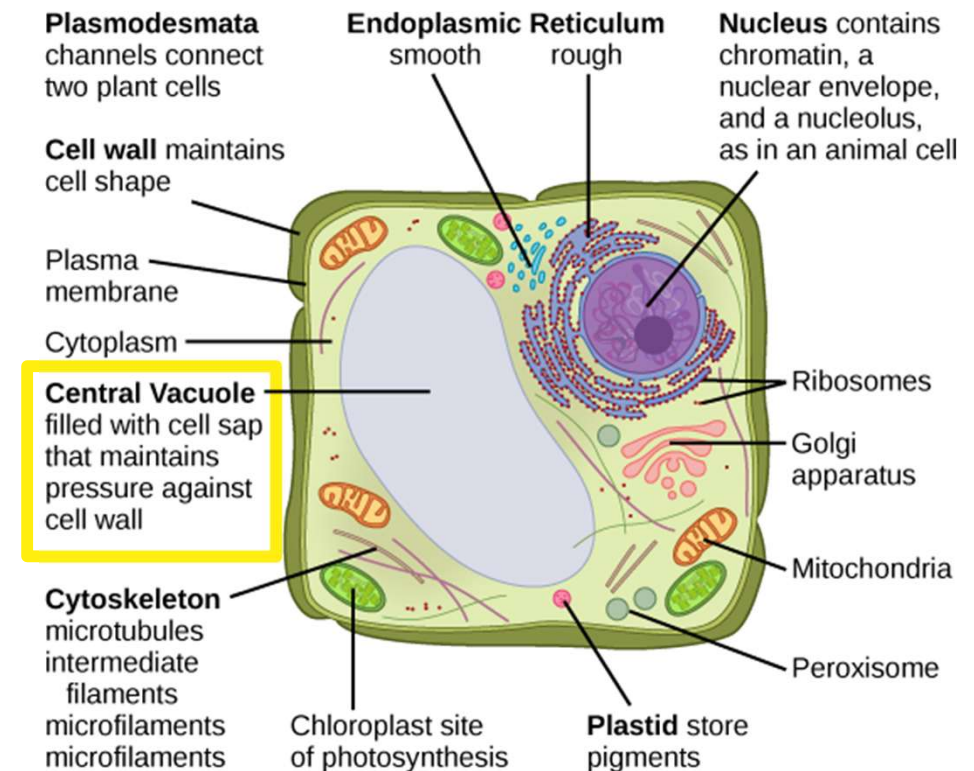
- **Chloroplasts** = site of photosynthesis

- contains pigments (usually green) that capture the energy of sunlight & store it in sugar

– more on this in a future chapter

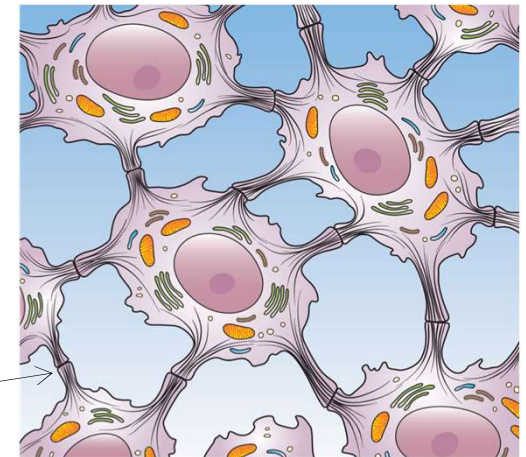


- Do plant cells have anything that animal cells don't?
- **Central vacuole:** very large – takes up half or more of a plant cell & has many functions
 - filled with fluid – helps maintain water balance
 - stores wastes & nutrients
 - contains pigments that give flowers their color
 - provide pressure “turgor pressure” to keep cells rigid



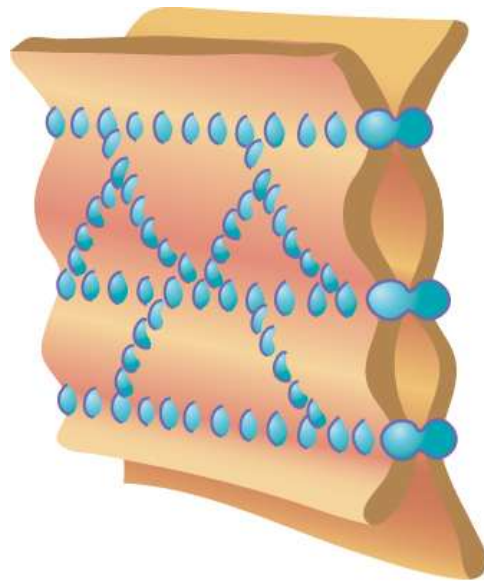
How do cells stick together & communicate?

- Single-celled organisms exist on their own
- But multi-cellular eukaryotic organisms (like plants & animals) have to stick together & communicate with each other in order to work as a unit

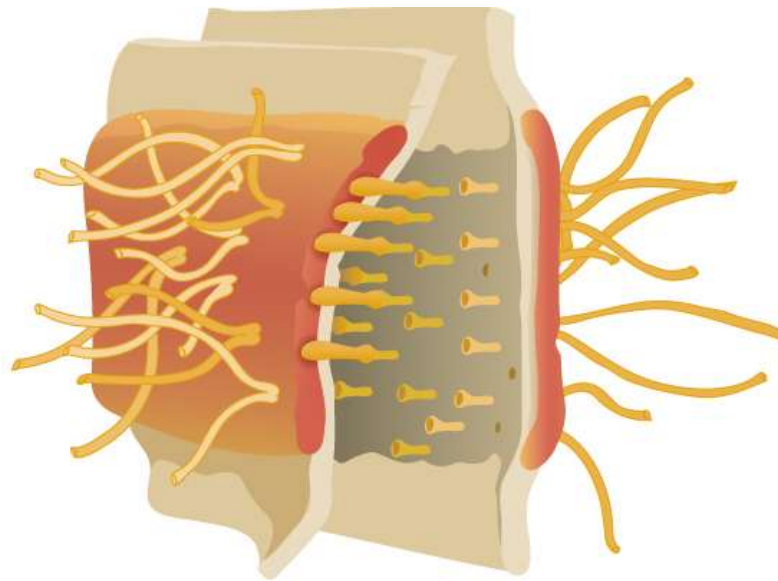


- **Cell junctions** allow for attachment & communication between cells
 - 3 types: desmosomes, tight junctions, & gap junctions (animals) or plasmodesmata (plants)

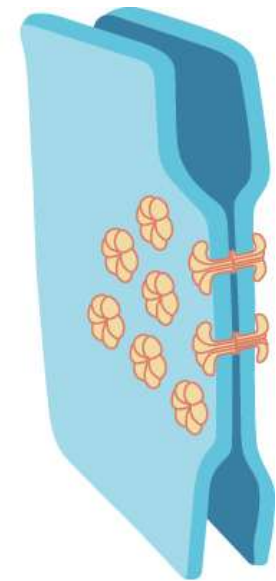
- **Desmosomes** attach cells together
 - Found where cells need to adhere tightly together under the stresses of movement
 - e.g. the skin, intestine, & bladder



tight junctions

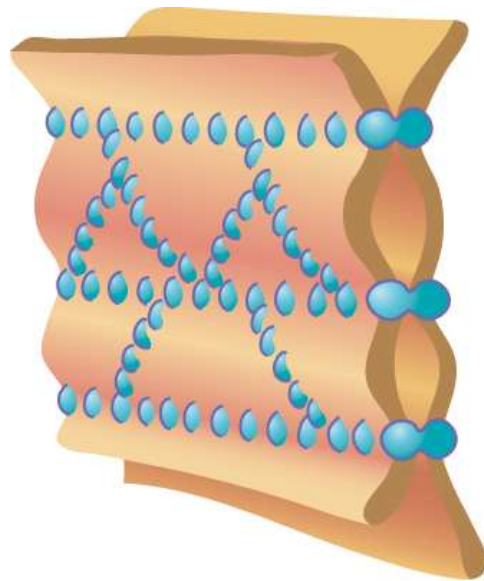


desmosomes

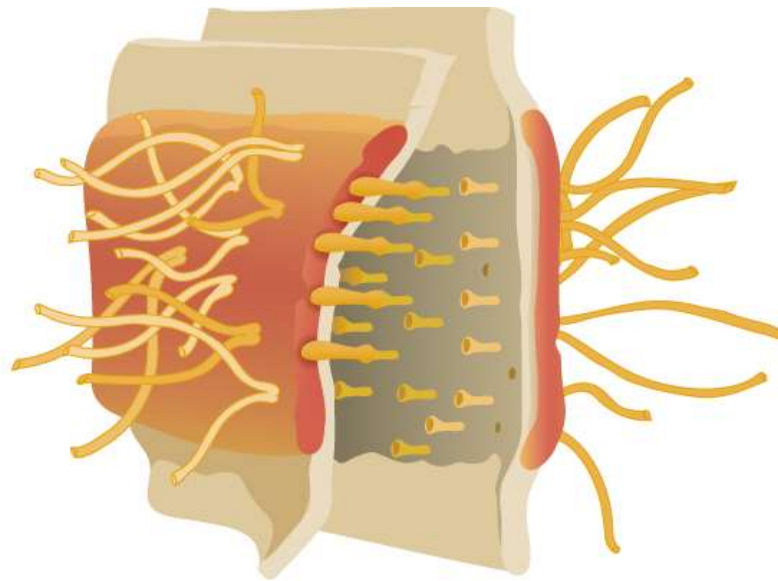


gap junctions

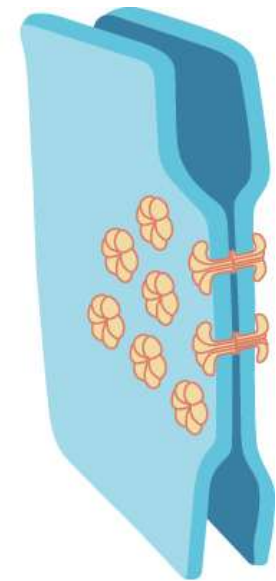
- **Tight Junctions** attach cells together & make them leak proof
 - Found where tubes & sacs must hold contents without leaking
 - e.g. the skin, bladder, & stomach



tight junctions

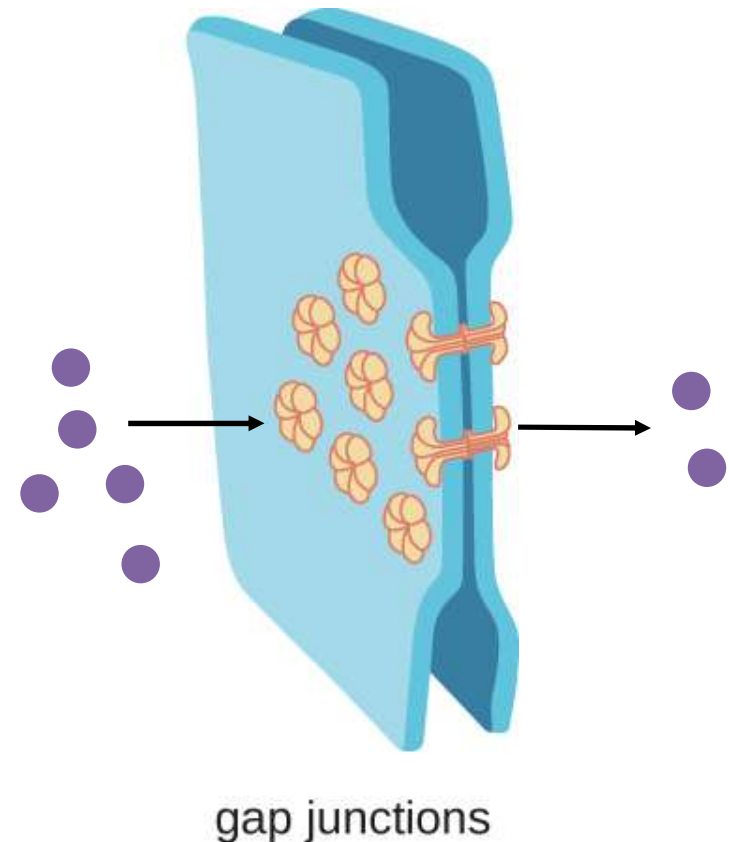


desmosomes

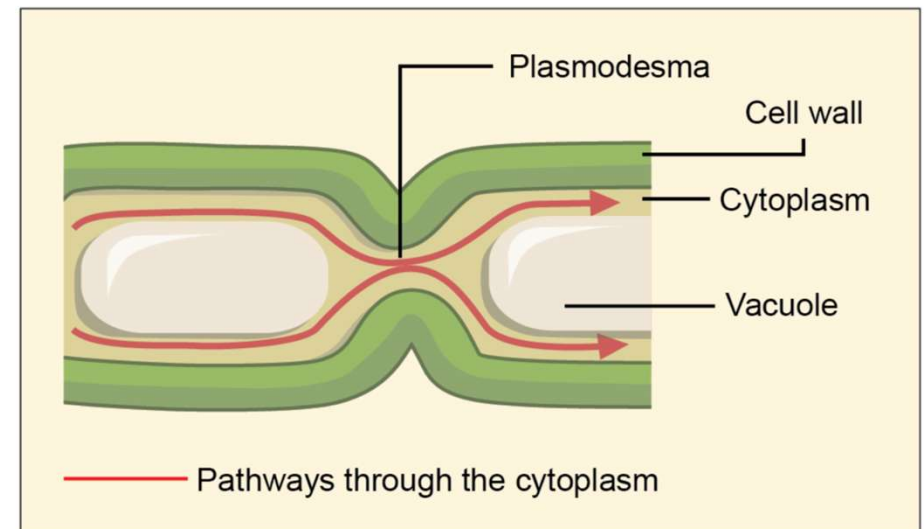
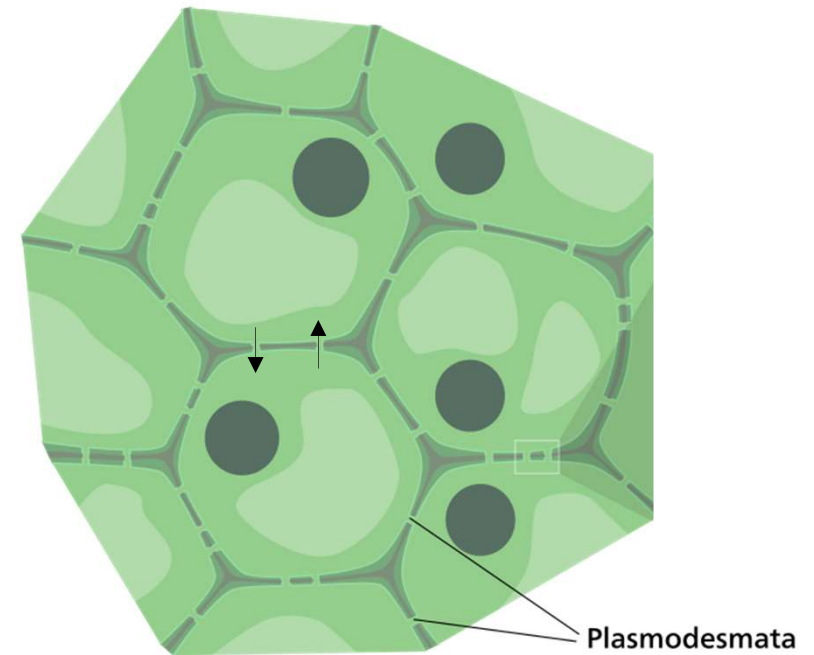


gap junctions

- **Gap Junctions** allow for communication between animal cells
 - allow passage of hormones, nutrients, & ions – this synchronizes cells
 - e.g. link heart cells together so that all contract at once
 - e.g. coordinate cells in digestive tract so that food is propelled in only one direction



- **Plasmodesmata** allow for communication between plant cells
 - Tunnels in the cell wall of plants allow hormones, nutrients, etc. to pass



- We now know:
 - cells have organelles & other structures that build materials (proteins, lipids, etc.)
 - some of these materials must leave the cell to get to the blood stream (e.g. hormones, insulin, or antibodies of the immune system)
 - cells have mitochondria: break apart glucose for energy
- So we must be able to get things (like glucose) into the cell & things (like antibodies) out of a cell
- BUT: we also learned that the cell membrane encloses the cell, keeping the insides in & the outsides out...

So how do we get things across the cell membrane?

- *Up next:*
 - **Cell membranes** – how do things cross?